

Application Layer

Vite / React frontend · FastAPI service (/predict /run-cgem /ood/score /sensitivity/sobol /sweep ...)

ML Extension Layer

XGBoost surrogate emulator · Mondrian conformal intervals · CQR (time-to-G-LOC) · Mahalanobis OOD · Sobol/Morris sensitivity

Python Wrapper

Input encoding · Subprocess orchestration (isolated tmpdir per call) · Output parsing · Batch + parallel support

FAA-validated CGEM Fortran Core

Fortran ODE solver (src/cgem.f) · Cardiovascular / cerebrovascular +Gz physiology · NOT modified

Data flow: bottom → top · Additive wrapper — CGEM core unchanged